Results in Control and Optimization 4 (2021) 100040

Contents lists available at ScienceDirect

Results in Control and Optimization

journal homepage: www.elsevier.com/locate/rico

A new score function based Fermatean fuzzy transportation problem

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ARTICLE INFO

Keywords:
Fermatean fuzzy numbers
Pythagorean fuzzy numbers
Transportation problem
Score function
Accuracy function
Excel solver

ABSTRACT

Today the supply, the demand and transportation costs in a transportation problem (TP) cannot be measured precisely due to the inconstant economic and environment conditions. The aim of this paper is to solve transportation problem where supply, demand and transportation costs are Fermatean fuzzy numbers (FFNs). In the existing literature, several methods have been proposed for solving transportation problems with fuzzy parameters (FPs) but in all the proposed methods the parameters relating to the transportation problems are either generalized fuzzy numbers or Pythagorean fuzzy numbers (PFNs). The concept of Fermatean fuzzy sets (FFSs) are very new and using Fermatean fuzzy sets one can handle uncertain information more easily in the process of decision making. Therefore, in this paper, first time we have solved transportation problem with Fermatean fuzzy parameters. For solving the transportation problem with Fermatean fuzzy parameters, we have proposed an algorithm and solved the same by existing method. Then using arithmetic operations of Fermatean fuzzy number the optimal value can be obtained. To illustrate the proposed method we have solved a numerical example and computed results are presented and compared with the existing literature. Finally, the significance of the paper and the scope of future study are mentioned.

1. Introduction

In the present circumstances, the pressure on industries/farms or a manufacturing companies to perceive most effective ways to produce and deliver goods to customers has become a formidable task. Again, continued economic deregulation, increased safety, increasing customer expectations, rising globalization, refined modern technological systems and continually changing mode of the transportation services, the manager of the transportation system faces a lot of difficulties for delivering goods to the customers. The time and place of delivering the products to the customers in terms of quantities, they want in a cost-effective manner, become more challenging. And that is why; transportation management is one of the key strategic decisions in modern supply chain business environment. Therefore, transportation problems (TPs) yield a best solution to achieve this goal. This transportation problem ensures the efficient movement and timely availability of raw materials and finished products. The TP is one of the most important mathematical programming problems which arise in many real-life decision making problems. Hence, TP has paying more attention in the literature of management or managerial decision making. The primary idea of TP is to hand out product/goods from certain places to some destinations in such a way that the total transportation cost is minimized. The TP has many real-life applications, viz. scheduling, routing, production planning, plant location problems, waste management, tourism management, inventory control and renewable energy location problem etc. For more details one may refer to the influential works of Tirkolaee et al. [1–6], Barman et al. [7], Goli et al. [8], Ozceylan and Paksoy [9], Paksoy et al. [10,11], Demirel et al. [12], Bhaumik et al. [13] and Lotfi et al. [14].

The classical TP was first introduced by Hitchcock [15] for the purpose of product distribution from several sources to numerous localities. The TPs can be modeled as a standard linear programming problem (LPP), which can be solved by the well-known simplex method. There are several methods for finding initial basic feasible solution for TP. These methods are north-west corner,

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https://doi.org/10.1016/j.rico.2021.100040
Received 9 April 2021; Received in revised form 13 June 2021; Accepted 7 July 2021
Available online 13 July 2021
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row minima, column minima, matrix minima and Vogel's approximation method. Korukoglu and Balli [16] introduced an improved Vogel's approximation method for transportation problem. Also, there are several evolutionary algorithms to solve the TP.

In traditional TPs, it has been generally assumed that decision-maker (DM) is sanguine about the precise values of transportation cost, supply and demand of the product. But, in fact all the parameters of the TPs may not be fixed /precise due to uncontrollable factors of dynamic economic environment. Therefore, parameters of the TPs are imprecise due to uncertainty and dearth of proper information. The uncertainty may occur due to some factors viz., (i) DM has no idea about the transportation cost, supply and demand when goods/products are shifted to be scheduled at the beginning time. As a result some uncertainty may occur in connection with the transportation cost, supply and demand (ii) now a days present market situation is always flooded i.e., always unstable due to competitive market and hence transportation cost, supply and demand are totally unpredictable. To combat these types of situation, the problem can be modeled using the concept of uncertainty and costs, supply and demands are treated in terms of imprecise values. In such cases fuzzy set (FS) theory plays an important role to handle such situation. Therefore, fuzzy decision making method is more important here. The concept of FS was introduced by Zadeh [17] in 1965 and it dealt with imprecision, vagueness in real world situations. In the year 1970, Belmann and Zadeh [18] introduced the concept of decision making problems involving uncertainty. Zimmermann [19] showed that the solutions obtained by fuzzy linear programming are always optimum and efficient. Chanas et al. [20] introduced a fuzzy linear programming model for solving TPs with precise costs, fuzzy supply and fuzzy demand. Chanas and Kuchta [21] proposed the idea of the optimal solution for the TP with fuzzy coefficients expressed in terms of fuzzy numbers, and developed an algorithm for finding the optimal solution. Since then several researchers have formulated the TPs either single objective optimization problem or multi-objective optimization problem by considering various fuzzy environments. They also have proposed several methods for finding the optimal solutions. In case of single objective optimization, one may refer to the works of Tada and Ishii [22], Hashmi et al. [23], Liu et al. [24], Kundu et al. [25], Singh and Yadav [26], Gupta and Kumari [27], Arora [28], Bharti and Singh [29], Kumar [30], Kaur and Kumar [31]. In case of multi-objective optimization, one may mention more dominant works of Midya et al. [32], Ghosh and Roy [33], Ghosh et al. [34], Das et al. [35], Das and Roy [36], Roy et al. [37], Ahmad and Adhami [38] and Li and Lai [39].

However, the FS takes only a membership function and the degree of non-membership function which is just a compliment of the degree of membership function. There may be a situation where the sum of the membership function and non-membership function is greater than one. Thus orthopair FSs have been introduced in which the membership grades of an element $x \in X$ are pairs of values in the unit interval $\langle \mu(x), \gamma(x) \rangle$, one of which indicates support of membership in the fuzzy set and other indicates support against membership. For example, Atanassov's classical intuitionistic fuzzy sets (IFSs) [40–42] and Atanassov's second kind of IFSs [43,44].

Recently, Yager [45,46] introduced another orthopair of FS, known as Pythagorean fuzzy set (PFS), where the square sum of the support of membership and support against membership value is equal to or less than one. PFSs have attracted the attention of many researchers within a short period of time and several researchers have also proposed real-life applications under Pythagorean fuzzy environment. For more details one may mention the works of Li et al. [47], Zhou et al. [48], Bolturk et al. [49], Qin [50], Wan et al. [51], Chen [52] and Kumar et al. [53]. But, if orthopair fuzzy set as <0.9, 0.6>, where 0.9 is the support of the membership of certain criteria of a parameter and 0.6 is the support against membership then it does not follow the condition of IFS as well as PFS. However, the cubic sum of the support of membership and support against membership degrees is equal to or less than one. And in this situation Senapati and Yager [54] very recently introduced Fermatean Fuzzy set (FFS). They also showed that FFSs have more uncertain than IFSs and PFSs and are capable of handling higher level of uncertainties [55].

Based on the earlier discussions on TP and recently available several research articles on TP, there are no existing methodologies which are available on TP under Fermatean fuzzy environment. Hence, there is an essential urgency to introduce a new solution methodology for solving TP in the light of Fermatean fuzzy environment. To the best of our knowledge and belief, there are no optimization models in the literate for TP in Fermatean fuzzy environment. For this fact, in this paper, we have treated cost parameters, demand and supply parameters of a TP as FFS as FFSs are most fruitful FS sets which are more competent to manage higher level of uncertainties.

In this paper, a method is proposed for solving TPs in Fermatean fuzzy environment. In the proposed method transportation cost, supply and demand of the product are represented by orthopair FS $\langle \alpha, \beta \rangle$ which satisfies the relation $0 \leq \alpha^3 + \beta^3 \leq 1$ and using score function and accuracy function of the FFSs we have converted the TP into crisp TPs. Then the converted TPs have been solved by excel solver to find out the optimal solutions. Solver is a Microsoft Excel add-in program for data analysis. Using solver one can determine the maximum or minimum value of one cell by changing other cells. The use of excel solver for analysis of operations research problems is an important tool and more convenient in contemporary technological world. It is laborious for a DM to solve optimization problems (linear or non-linear) manually when decision making problems may occur with large number of decision variables. A work that can take several days or weeks to solve manually, could be done in seconds using excel solver. Finally, to illustrate the proposed method we have solved a numerical example and computed results are presented and compared with the existing literature. All the requisite abbreviations are given in Table 1. Table 2 presents the significance influences of the different authors towards TP under various fuzzy environments. Significance influences of the different authors towards real-life decision making problem are listed in Table 3.

The main contributions of this research work are as follows:

(i) In FFTP, all the parameters viz. transportation cost, supply and demand are considered as FFSs.

(ii) Three newly score functions are proposed for the ranking of FFSs.

(iii) Ranking/order relation of two FFSs are proposed.

Table 1
List of abbreviations.

Abbreviations	Full name
DM	Decision-maker
FFN	Fermatean fuzzy number
FFS	Fermatean fuzzy set
FFTP	Fermatean fuzzy transportation problem
FS	Fuzzy set
FTP	Fuzzy transportation problem
IFS	Intuitionistic fuzzy set
LPP	Linear programming problem
PFS	Pythagorean fuzzy set
PFTP	Pythagorean fuzzy transportation problem
TP	Transportation problem
FP	Fuzzy parameter
FN	Fuzzy number

Table 2
Significance influences of the different authors towards TP under various fuzzy environment.

References	Year	Significance influences
Chanas et al. [20]	1984	Fuzzy approach to the transportation problem
Chanas and Kuchata [21]	1996	Concept of the optimal solution of the transportation problem with fuzzy cost coefficients
Tada and Ishii [22]	1996	Integer fuzzy transportation problem
Li and Lai [39]	2000	Fuzzy approach to the multi-objective transportation problem
Liu and Kao [56]	2004	Fuzzy transportation problems based on extension principle
Dinagar and Palanivel [57]	2009	Transportation problem in fuzzy environment
Pandian and Natarajan [58]	2010	Algorithm for finding a fuzzy optimal solution for fuzzy transportation problems
Kaur and Kumar [31]	2011	Fuzzy transportation problems using ranking function
Kundu et al. [25]	2014	Fixed charge transportation problem with type-2 fuzzy variables
Liu et al [24]	2014	Solid transportation problem with type-2 fuzzy variables
Singh and Yadav [26]	2016	Solving intuitionistic fuzzy transportation problem of type-2
Gupta and Kumari [27]	2017	Method for solving intuitionistic fuzzy transportation problem of type-2
Arora [28]	2018	Interval-valued fuzzy fractional transportation problem
Bharti and Singh [29]	2018	Transportation problem under interval-valued intuitionistic fuzzy environment
Kumar [30]	2018	Intuitionistic fuzzy solid transportation problems
Ahmad and Adhami [38]	2019	Neutrosophic programming approach to multi-objective nonlinear transportation problem with fuzzy parameters
Hashmi et al. [23]	2019	Two stage fixed charge transportation problem with multiple objectives and fuzzy linguistic preferences
Kumar et al. [53]	2019	Pythagorean fuzzy approach to the transportation problem
Roy et al. [37]	2019	Multi-objective fixed-charge transportation problem under two-fold uncertainty
Das and Roy [36]	2019	Effect of variable carbon emission in a multi-objective transportation-p-facility location problem under neutrosophic environment
Das et al. [35]	2020	Application of type-2 fuzzy logic to a multi-objective green solid transportation-location problem with dwell time under carbon tax, cap and offset policy: Fuzzy versus Non-fuzzy techniques,
Ghosh et al. [34]	2021	Multi-objective fully intuitionistic fuzzy fixed-charge solid transportation problem
Ghosh and Roy [33]	2021	Fuzzy-rough multi-objective product blending fixed-charge transportation problem with truck load constraints through transfer station
Midya et al. [32]	2021	Intuitionistic fuzzy multi-stage multi-objective fixed-charge solid transportation problem in green supply chain

(iv) A solution methodology is proposed to solve FFTP.
(v) Finding optimal solution, we use Excel solver.

The rest of the paper is organized as follows: in Section 2 some basic definitions about PFSs and FFSs are presented. The mathematical formulation of TP is presented in Section 3. In Section 4, proposed solution methodology is discussed. The results of Fermatean fuzzy transportation problem (FFTP) are discussed in Section 5. The conclusions are drawn in Section 6.

2. Preliminaries

In this section, some fundamental definitions about PFSs and FFSs are presented. Also, we have proposed some score functions to accomplish the entire paper.

Definition 2.1 ([46]). Let X be a Universal set. A PFS is an object of the form $\widetilde{P} = \left\{\left\langle x, \alpha_{\widetilde{P}}(x), \beta_{\widetilde{P}}(x)\right\rangle : x \in X\right\}$ where $\alpha_{\widetilde{P}}(x) : X \to [0,1]$ and $\beta_{\widetilde{P}}(x) : X \to [0,1]$ which satisfies the relation $0 \le (\alpha_{\widetilde{P}}(x))^2 + (\beta_{\widetilde{P}}(x))^2 \le 1, \forall x \in X$. The values $\alpha_{\widetilde{P}}(x)$ and $\beta_{\widetilde{P}}(x)$ denote, respectively, the degree of membership and the degree of non-membership of the element $x \in X$ in the PFS $\widetilde{P}$.

For any PFS $\widetilde{P}$ and $x \in X$, the degree of indeterminacy is represented by $\pi_P(x) = \sqrt{1 - (\alpha_{\widetilde{P}}(x))^2 - (\beta_{\widetilde{P}}(x))^2}$. For simplicity, we shall denote the object $\widetilde{P} = \left\{\left\langle x, \alpha_{\widetilde{P}}(x), \beta_{\widetilde{P}}(x)\right\rangle : x \in X\right\}$ in the form $\widetilde{P} = \left\langle \alpha_{\widetilde{P}}, \beta_{\widetilde{P}}\right\rangle$.

Table 3
Significance influences of the different authors towards real-life decision making problem.

References	Year	Significance influences
Paksoy et al. [10]	2012	Fuzzy multi-objective optimization of a green supply chain network with risk management that includes environmental hazards
Paksoy et al. [11]	2013	Mixed model disassembly line balancing problem with fuzzy goals
Yager [46]	2014	Pythagorean membership grades in multi-criteria decision making
Azevedo et al. [59]	2014	Dynamic programming for a Markov-switching jump–diffusion
Ozceylan and Paksoy [9]	2014	Interactive fuzzy programming approaches to the strategic and tactical planning of a closed-loop supply chain under uncertainty
Demirel et al. [12]	2014	A genetic algorithm approach for optimizing a closed-loop supply chain network with crisp and fuzzy objectives
Savku et al. [60]	2014	Optimal control of stochastic hybrid models in the framework of regime switches
Garg [61]	2017	Interval valued Pythagorean fuzzy set based TOPSIS method
Geng et al. [62]	2017	Multiple criteria group decision making using Pythagorean fuzzy uncertain linguistic TODIM method
Tirkolaee et al. [5]	2017	A robust multi-trip vehicle routing problem of perishable products with intermediate depots and time windows
Boltruk [49]	2018	Supplier selection in a manufacturing firm using Pythagorean fuzzy
Chen T-Y [52]	2018	Risk attitudinal assignment model involving Pythagorean fuzzy information and its application to financial decision making
Garg [63]	2018	Interval valued Pythagorean fuzzy numbers and its application to decision-making
Qin [50]	2018	Multiple attribute sir group decision model using generalized Pythagorean fuzzy Maclaurin symmetric means
Kropat and Weber [64]	2018	Fuzzy target-environment networks and fuzzy-regression approaches
Senapati and Yager [55]	2019	Multi-criteria decision-making using Fermatean fuzzy weighted averaging/geometric operators
Tirkolaee et al. [2]	2019	A robust bi-objective multi-trip periodic capacitated arc routing problem for urban waste collection using a multi-objective invasive weed optimization
Tirkolaee et al. [3]	2019	A hybrid augmented ant colony optimization for the multi-trip capacitated arc routing problem under fuzzy demands for urban solid waste management
Tirkolaee et al. [4]	2019	Multi-objective aggregate production planning model considering overtime and out sourcing options under fuzzy seasonal demand
Tirkolaee et al. [1]	2020	Just-in-time energy-aware flow shop scheduling problem with outsourcing option using self-adaptive artificial fish swarm algorithm
Lotfi et al. [14]	2021	Robust bi-level programming for renewable energy location
Barman et al. [7]	2021	Back-ordered inventory model with inflation in a cloudy-fuzzy environment
Bhaumik et al. [13]	2021	Multi-objective linguistic-neutrosophic matrix game and its applications to tourism management
Tirkolaee et al. [6]	2021	Sustainable fuzzy multi-trip location-routing problem for medical waste management during the COVID-19 outbreak
Goli et al. [8]	2021	Fuzzy integrated cell formation and production scheduling considering automated guided vehicles and human factors
Sahoo [65]	2021	Some Score Functions on Fermatean Fuzzy Sets and Its Application to Bride Selection Based on TOPSIS Method

Definition 2.2 ([46]). Let $\widetilde{P} = \langle \alpha_{\widetilde{P}}, \beta_{\widetilde{P}} \rangle$ be any PFS, then score function of $\widetilde{P}$ denoted by $S_P(\widetilde{P})$ and is defined as $S_P(\widetilde{P}) = \frac{1}{2}(1 + \alpha^2_{\widetilde{P}} - \beta^2_{\widetilde{P}})$.

Definition 2.3 ([46]). Let $\widetilde{P} = \langle \alpha_{\widetilde{P}}, \beta_{\widetilde{P}} \rangle$ be any PFS, then accuracy function of $\widetilde{P}$ denoted by $H_P(\widetilde{P})$ and is defined as $H_P(\widetilde{P}) = \alpha^2_{\widetilde{P}} + \beta^2_{\widetilde{P}}$.

Definition 2.4 ([54]). Let X be a Universal set. A FFS is an object of the form $\widetilde{F} = \{\langle x, \alpha_{\widetilde{F}}(x), \beta_{\widetilde{F}}(x) \rangle : x \in X\}$ where $\alpha_{\widetilde{F}}(x) : X \to [0,1]$ and $\beta_{\widetilde{F}}(x) : X \to [0,1]$ which satisfies the relation $0 \le (\alpha_{\widetilde{F}}(x))^3 + (\beta_{\widetilde{F}}(x))^3 \le 1, \forall x \in X$. The values $\alpha_{\widetilde{F}}(x)$ and $\beta_{\widetilde{F}}(x)$ denote, respectively, the degree of membership and the degree of non-membership of the element $x \in X$ in the set $\widetilde{F}$.

For any FFS $\widetilde{F}$ and $x \in X$, the degree of indeterminacy is represented by $\pi_F(x) = \sqrt[3]{1 - (\alpha_{\widetilde{F}}(x))^3 - (\beta_{\widetilde{F}}(x))^3}$. For simplicity, we shall denote the object $\widetilde{F} = \{\langle x, \alpha_{\widetilde{F}}(x), \beta_{\widetilde{F}}(x) \rangle : x \in X\}$ in the form $\widetilde{F} = \langle \alpha_{\widetilde{F}}, \beta_{\widetilde{F}} \rangle$.

Definition 2.5 ([54]). Let $\widetilde{F} = \langle \alpha_{\widetilde{F}}, \beta_{\widetilde{F}} \rangle$, $\widetilde{F}_1 = \left\langle \alpha_{\widetilde{F}_1}, \beta_{\widetilde{F}_1} \right\rangle$ and $\widetilde{F}_2 = \left\langle \alpha_{\widetilde{F}_2}, \beta_{\widetilde{F}_2} \right\rangle$ be three FFSs over the universe X and $\lambda > 0$ then their basic arithmetical operations are defined as follows:

(i) Addition: $\widetilde{F}_1 \oplus \widetilde{F}_2 = \left\langle \sqrt[3]{(\alpha_{\widetilde{F}_1})^3 + (\alpha_{\widetilde{F}_2})^3 - (\alpha_{\widetilde{F}_1})^3(\alpha_{\widetilde{F}_2})^3}, \beta_{\widetilde{F}_1}\beta_{\widetilde{F}_2} \right\rangle$

(ii) Multiplication: $\widetilde{F}_1 \otimes \widetilde{F}_2 = \left\langle \alpha_{\widetilde{F}_1}\alpha_{\widetilde{F}_2}, \sqrt[3]{(\beta_{\widetilde{F}_1})^3 + (\beta_{\widetilde{F}_2})^3 - (\beta_{\widetilde{F}_1})^3(\beta_{\widetilde{F}_2})^3} \right\rangle$

(iii) Scalar multiplication: $\lambda \odot \widetilde{F} = \left\langle \sqrt[3]{1 - (1 - (\alpha_{\widetilde{F}})^3)^\lambda}, (\beta_{\widetilde{F}})^\lambda \right\rangle$

(iv) Exponent: $\widetilde{F}^\lambda = \left\langle (\alpha_{\widetilde{F}})^\lambda, \sqrt[3]{1 - (1 - (\beta_{\widetilde{F}})^3)^\lambda} \right\rangle$

Definition 2.6 ([54]). Let $\widetilde{F}_1 = \left\langle \alpha_{\widetilde{F}_1}, \beta_{\widetilde{F}_1} \right\rangle$ and $\widetilde{F}_2 = \left\langle \alpha_{\widetilde{F}_2}, \beta_{\widetilde{F}_2} \right\rangle$ be two FFSs over the universe X, then their set operations are defined as follows:
(i) Union: $\widetilde{F}_1 \cup \widetilde{F}_2 = \left\langle \max(\alpha_{\widetilde{F}_1} \alpha_{\widetilde{F}_2}), \min(\beta_{\widetilde{F}_1} \beta_{\widetilde{F}_2}) \right\rangle$
(ii) Intersection: $\widetilde{F}_1 \cap \widetilde{F}_2 = \left\langle \min(\alpha_{\widetilde{F}_1} \alpha_{\widetilde{F}_2}), \max(\beta_{\widetilde{F}_1} \beta_{\widetilde{F}_2}) \right\rangle$
(iii) Complement: $(\widetilde{F}_1)^c = \left\langle \beta_{\widetilde{F}_1}, \alpha_{\widetilde{F}_1} \right\rangle$

For more details about FFSs one may refer to the works of Senapati and Yager [54,55].

Definition 2.7 (*Senapati and Yager [54,55]*). Let $\widetilde{F} = \left\langle \alpha_{\widetilde{F}}, \beta_{\widetilde{F}} \right\rangle$ be any FFS, then score function of $\widetilde{F}$ denoted by $S_F(\widetilde{F})$ and is defined as $S_F(\widetilde{F}) = (\alpha^3_{\widetilde{F}} - \beta^3_{\widetilde{F}})$.

The score function defined Senapati and Yager [54,55], the score value lies in the interval [−1,1] i.e. $S_F(\widetilde{F}) \in [-1, 1]$. It is to be noted that the function is positive when $S_F(\widetilde{F}) \in [0, 1]$ and negative when $S_F(\widetilde{F}) \in [-1, 0)$.

In the purpose of ranking of FNs/ FSs (either IFS or PFS) most of the researchers have considered score functions whose score values are lies in the interval 0 and 1. So to maintain the same we have proposed some score functions for the ranking of FFSs. For more details about the same one may refer to the work of Sahoo [65].
(i) (***Type 1***) $S_{1F}(\widetilde{F}) = \frac{1}{2}(1 + \alpha^3_{\widetilde{F}} - \beta^3_{\widetilde{F}})$
(ii) (***Type 2***) $S_{2F}(\widetilde{F}) = \frac{1}{3}(1 + 2\alpha^3_{\widetilde{F}} - \beta^3_{\widetilde{F}})$
(iii) (***Type 3***) $S_{3F}(\widetilde{F}) = \frac{1}{2}(1 + \alpha^2_{\widetilde{F}} - \beta^2_{\widetilde{F}}) \left|\alpha_{\widetilde{F}} - \beta_{\widetilde{F}}\right|$

Property 2.1. *Let $\widetilde{F} = \left\langle \alpha_{\widetilde{F}}, \beta_{\widetilde{F}} \right\rangle$ be any FFS then $S_{1F}(\widetilde{F}) \in [0, 1]$, $S_{2F}(\widetilde{F}) \in [0, 1]$ and $S_{2F}(\widetilde{F}) \in [0, 1]$.*

Proof. For any FFS $\alpha^3_{\widetilde{F}} \geq 0$ and $\beta^3_{\widetilde{F}} \leq 1 \Rightarrow 1 - \beta^3_{\widetilde{F}} \geq 0$. Hence, $1 + \alpha^3_{\widetilde{F}} - \beta^3_{\widetilde{F}} \geq 0$ and obviously $S_{1F}(\widetilde{F}) \geq 0$. Again, $\alpha^3_{\widetilde{F}} + \beta^3_{\widetilde{F}} \leq 1 \Rightarrow 1 + \alpha^3_{\widetilde{F}} + \beta^3_{\widetilde{F}} \leq 2 \Rightarrow 1 + \alpha^3_{\widetilde{F}} - \beta^3_{\widetilde{F}} \leq 2$ as $\beta^3_{\widetilde{F}} \geq 0$. So $\frac{1+\alpha^3_{\widetilde{F}} - \beta^3_{\widetilde{F}}}{2} \leq 1$ and $S_{1F}(\widetilde{F}) \leq 1$.

Hence for Type 1 score function $S_{1F}(\widetilde{F}) \in [0, 1]$. In similar manner one can prove that $S_{2F}(\widetilde{F}) \in [0, 1]$ and $S_{3F}(\widetilde{F}) \in [0, 1]$. Particularly, if $\widetilde{F} = \langle 0, 1 \rangle$, then $S_{1F}(\widetilde{F}) = 0$, $S_{2F}(\widetilde{F}) = 0$ and $S_{3F}(\widetilde{F}) = 0$. Again if $\widetilde{F} = \langle 1, 0 \rangle$, then $S_{1F}(\widetilde{F}) = 1$, $S_{2F}(\widetilde{F}) = 1$ and $S_{3F}(\widetilde{F}) = 1$

Definition 2.8 ([54]). Let $\widetilde{F} = \left\langle \alpha_{\widetilde{F}}, \beta_{\widetilde{F}} \right\rangle$ be any FFS then accuracy function of $\widetilde{F}$ denoted by $H_F(\widetilde{F})$ and is defined by $H_F(\widetilde{F}) = \alpha^3_{\widetilde{F}} + \beta^3_{\widetilde{F}}$.

Definition 2.9. Let $\widetilde{F}_1 = \left\langle \alpha_{\widetilde{F}_1}, \beta_{\widetilde{F}_1} \right\rangle$ and $\widetilde{F}_2 = \left\langle \alpha_{\widetilde{F}_2}, \beta_{\widetilde{F}_2} \right\rangle$ be two FFSs. Then ranking or order relations of $\widetilde{F}_1$ and $\widetilde{F}_2$ are defined as follows:
(i) $\widetilde{F}_1 > \widetilde{F}_2$ iff either $(S_{kF}(\widetilde{F}_1) > S_{kF}(\widetilde{F}_2))$ or $(S_{kF}(\widetilde{F}_1) = S_{kF}(\widetilde{F}_2)$ and $H_F(\widetilde{F}_1) > H_F(\widetilde{F}_2))$
(ii) $\widetilde{F}_1 < \widetilde{F}_2$ iff either $(S_{kF}(\widetilde{F}_1) < S_{kF}(\widetilde{F}_2))$ or $(S_{kF}(\widetilde{F}_1) = S_{kF}(\widetilde{F}_2)$ and $H_F(\widetilde{F}_1) < H_F(\widetilde{F}_2))$
(iii) $\widetilde{F}_1 = \widetilde{F}_2$ iff $S_{kF}(\widetilde{F}_1) = S_{kF}(\widetilde{F}_2)$ and $H_F(\widetilde{F}_2) = H_F(\widetilde{F}_2)$

Example 2.1 (*Senapati and Yager [54,55]*). If $\widetilde{F}_1 = \langle 0.93, 0.60 \rangle$ and $\widetilde{F}_2 = \langle 0.85, 0.75 \rangle$ then we have the following:
For Type 1: $S_{1F}(\widetilde{F}_1) = 0.7565$ and $S_{1F}(\widetilde{F}_2) = 0.5961$ and hence, $S_{1F}(\widetilde{F}_1) > S_{1F}(\widetilde{F}_2) \Rightarrow \widetilde{F}_1 > \widetilde{F}_2$.
For Type 2: $S_{2F}(\widetilde{F}_1) = 0.7473$ and $S_{2F}(\widetilde{F}_2) = 0.6021$ and hence, $S_{2F}(\widetilde{F}_1) > S_{2F}(\widetilde{F}_2) \Rightarrow \widetilde{F}_1 > \widetilde{F}_2$.
For Type 3: $S_{3F}(\widetilde{F}_1) = 0.2175$ and $S_{3F}(\widetilde{F}_2) = 0.0580$ and hence, $S_{3F}(\widetilde{F}_1) > S_{3F}(\widetilde{F}_2) \Rightarrow \widetilde{F}_1 > \widetilde{F}_2$.

So we claim that our proposed score functions defined here are justified as all score functions gives the same results similar to Senapati and Yager [54,55].

3. Model formulation

The basic objective of transportation problem is to transport any item/product from m origins to n destinations, so that the total transportation cost is minimized. Suppose a farm/company has m warehouses (origins) and n outlets (destinations). A single item/product is to be shifted from the warehouses to the outlets. Also assumed that each warehouse has a given amount of supply and each outlet has a given amount of demand. Here, we have also assumed that the transportation cost of a unit measure of items/products from ith origins to jth destination is c_{ij}, the supply at ith origin is a_i and the demand of at jth destination is b_j and x_{ij} is the number of unit item/product shifted from ith origins to jth destination.

Mathematically, the transportation problem can be formulated as follows:

$$\text{minimize } z_0 = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} \tag{1}$$

subject to

$$\sum_{j=1}^{n} x_{ij} \leq a_i, \ i = 1, 2, \ldots, m$$

$$\sum_{i=1}^{m} x_{ij} \geq b_j,\ j = 1, 2, \ldots, n$$
$$x_{ij} \geq 0,\ i = 1, 2, \ldots, m,\ j = 1, 2, \ldots, n$$

In this problem (1), the DM has the right to utilizing precise parametric values. But in reality, the DM may not be able to set precise data/parametric values due to uncontrollable economic environment. Hence, in this problem, all the parameters associated with TP are considered to be FFSs. And the FFSs are of the form $\langle \alpha, \beta \rangle$, where $0 \leq \alpha^3 + \beta^3 \leq 1$ $0 \leq \alpha^3 + \beta^3 \leq 1$. Introducing impreciseness/fuzziness in traditional TP, FFTP is formulated as follows:

$$\text{minimize } \left\langle \alpha_{\tilde{z}_0}, \beta_{\tilde{z}_0} \right\rangle = \sum_{i=1}^{m} \sum_{j=1}^{n} \left\langle \alpha_{\tilde{c}_{ij}}, \beta_{\tilde{c}_{ij}} \right\rangle \odot x_{ij} \tag{2}$$

subject to

$$\sum_{j=1}^{n} x_{ij} \leq \left\langle \alpha_{\tilde{a}_i}, \beta_{\tilde{a}_i} \right\rangle,\ i = 1, 2, \ldots, m$$
$$\sum_{i=1}^{m} x_{ij} \geq \left\langle \alpha_{\tilde{b}_j}, \beta_{\tilde{b}_j} \right\rangle,\ j = 1, 2, \ldots, n$$

where $0 \leq \left(\alpha_{\tilde{z}_0}\right)^3 + \left(\beta_{\tilde{z}_0}\right)^3 \leq 1$

$$0 \leq (\alpha_{\tilde{a}_i})^3 + (\beta_{\tilde{a}_i})^3 \leq 1,\ i = 1, 2, \ldots, m$$
$$0 \leq (\alpha_{\tilde{b}_j})^3 + (\beta_{\tilde{b}_j})^3 \leq 1, j = 1, 2, \ldots, n$$
$$x_{ij} \geq 0 \text{ and } 0 \leq (\alpha_{\tilde{c}_{ij}})^3 + (\beta_{\tilde{c}_{ij}})^3 \leq 1,\ i = 1, 2, \ldots, m;\ j = 1, 2, \ldots, n$$

where $\left\langle \alpha_{\tilde{a}_i}, \beta_{\tilde{a}_i} \right\rangle$ is the total Fermatean fuzzy availability of the item/product at ith origin, $\left\langle \alpha_{\tilde{b}_j}, \beta_{\tilde{b}_j} \right\rangle$ is the total Fermatean fuzzy demand of the item/product at jth destination, $\left\langle \alpha_{\tilde{c}_{ij}}, \beta_{\tilde{c}_{ij}} \right\rangle$ is unit Fermatean fuzzy transportation cost from ith origin to jth destination.

Now, problem (2) is a mathematical model of a TP in Fermatean fuzzy environment, called as FFTP. It is to be noted that if $\sum_{i=1}^{m} \oplus \left\langle \alpha_{\tilde{a}_i}, \beta_{\tilde{a}_i} \right\rangle = \sum_{j=1}^{n} \oplus \left\langle \alpha_{\tilde{b}_j}, \beta_{\tilde{b}_j} \right\rangle$, then the FFTP is said to be balanced, otherwise it is called unbalanced FFTP. The symbol $\sum \oplus$ denoted as summation in terms of Fermatean addition sense.

4. Solution methodology for solving FFTP

There are several methods available in the existing literature for finding initial basic feasible solution and optimal solution for TP in different fuzzy environment viz. Intuitionistic fuzzy, Pythagorean fuzzy etc. These methods are north-west corner, row minima, column minima, matrix minima, Vogel's approximation method and linear programming technique. Also, there are several evolutionary algorithms to solve the same. However, none of them have considered the TP in Fermatean fuzzy environment, where all the parameters are FFSs. So, in this paper we have solved the TP in Fermatean fuzzy environment by using the propose algorithm and use of well-known excel solver. The different steps of the proposed algorithm are as follows:

Step 1: *For* $k = 1, 2, 3$, *calculate the following:*

$$S_{kF}(\left\langle \alpha_{\tilde{a}_i}, \beta_{\tilde{a}_i} \right\rangle),\ i = 1, 2, ..m$$

$$S_{kF}(\left\langle \alpha_{\tilde{b}_j}, \beta_{\tilde{b}_j} \right\rangle),\ j = 1, 2, ..n$$

and $S_{kF} \left\langle \alpha_{\tilde{c}_{ij}}, \beta_{\tilde{c}_{ij}} \right\rangle,\ i = 1, 2, \ldots, m; j = 1, 2, \ldots, n$

Step 2: *For* $k = 1, 2, 3$, *calculate the following:*

$$S_{kF}(\sum_{i=1}^{m} \oplus \left\langle \alpha_{\tilde{a}_i}, \beta_{\tilde{a}_i} \right\rangle)$$

$$\textit{and } S_{kF}(\sum_{j=1}^{n} \oplus \left\langle \alpha_{\tilde{b}_j}, \beta_{\tilde{b}_j} \right\rangle)$$

Step 3: *If* $S_{kF}(\sum_{i=1}^{m} \oplus \left\langle \alpha_{\tilde{a}_i}, \beta_{\tilde{a}_i} \right\rangle) = S_{kF}(\sum_{j=1}^{n} \oplus \left\langle \alpha_{\tilde{b}_j}, \beta_{\tilde{b}_j} \right\rangle)$, $k = 1, 2, 3$ *then the TP is balance TP and go to Step 5; otherwise go to Step 4.*

Step 4: *If* $S_{kF}(\sum_{i=1}^{m} \oplus \left\langle \alpha_{\tilde{a}_i}, \beta_{\tilde{a}_i} \right\rangle) \neq S_{kF}(\sum_{j=1}^{n} \oplus \left\langle \alpha_{\tilde{b}_j}, \beta_{\tilde{b}_j} \right\rangle)$, $k = 1, 2, 3$ *then introduce a dummy origin(s) or a dummy destination(s) (as the case may be) with zero transportation cost and provide excess availability or the requirement to make it balance one and go to Step 5.*

Step 5: *Formulate the balanced TP as LPP, and solve using excel solver to find optimum solution.*

It is to be noted that for solving the Pythagorean transportation problem by our proposed approach, one may replace the score function $S_{kF}(.)$ by $S_P(.)$.

Table 4
Input Data for transportation cost, supply and demand.

	O1	O2	O3	O4	Supply
WH1	$\langle 0.1, 0.9\rangle$	$\langle 0.2, 0.8\rangle$	$\langle 0.1, 0.8\rangle$	$\langle 0.1, 0.9\rangle$	$\langle 0.7, 0.1\rangle$
WH2	$\langle 0.01, 0.99\rangle$	$\langle 0.3, 0.9\rangle$	$\langle 0.3, 0.8\rangle$	$\langle 0.1, 0.7\rangle$	$\langle 0.8, 0.1\rangle$
WH3	$\langle 0.1, 0.8\rangle$	$\langle 0.4, 0.8\rangle$	$\langle 0.4, 0.9\rangle$	$\langle 0.2, 0.9\rangle$	$\langle 0.9, 0.1\rangle$
Demand	$\langle 0.4, 0.7\rangle$	$\langle 0.7, 0.3\rangle$	$\langle 0.8, 0.1\rangle$	$\langle 0.60832, 0.4\rangle$	

5. Result and discussion with a numerical example

In this section, we illustrate our proposed solution methodology with a suitable example. Consider a FFTP, where all the parameters are given in Table 4. In this problem, supplies $\left\langle \alpha_{\widetilde{a}_i}, \beta_{\widetilde{a}_i} \right\rangle, i = 1,2,3$, demands $\left\langle \alpha_{\widetilde{b}_j}, \beta_{\widetilde{b}_j} \right\rangle, j = 1,2,3,4$ and costs $\left\langle \alpha_{\widetilde{c}_{ij}}, \beta_{\widetilde{c}_{ij}} \right\rangle, i = 1,2,3;\ j = 1,2,3,4$ are considered as FFSs/FFNs. Here, WH1, WH2 and WH3 are three warehouses/origins and O1, O2, O3 and O4 are four outlets/destinations.

To solve the FFTP, we have solved the following LPP (3) where all the input data are given in Table 4.

$$\text{minimize } S_{kF}\left\langle \alpha_{z_0}, \beta_{\widetilde{z}_0} \right\rangle = \sum_{i=1}^{3}\sum_{j=1}^{4} S_{kF}(\left\langle \alpha_{\widetilde{c}_{ij}}, \beta_{\widetilde{c}_{ij}} \right\rangle) \odot x_{ij} \tag{3}$$

subject to

$$\sum_{j=1}^{4} x_{ij} \le S_{kF}(\left\langle \alpha_{\widetilde{a}_i}, \beta_{\widetilde{a}_i} \right\rangle),\ i = 1,2,3$$
$$\sum_{i=1}^{3} x_{ij} \ge S_{kF}(\left\langle \alpha_{\widetilde{b}_j}, \beta_{\widetilde{b}_j} \right\rangle),\ j = 1,2,3,4$$
$$\text{where } 0 \le \left(\alpha_{\widetilde{z}_0}\right)^3 + \left(\beta_{\widetilde{z}_0}\right)^3 \le 1$$
$$0 \le (\alpha_{\widetilde{a}_i})^3 + (\beta_{\widetilde{a}_i})^3 \le 1, i = 1,2,3$$
$$0 \le (\alpha_{\widetilde{b}_j})^3 + (\beta_{\widetilde{b}_j})^3 \le 1, j = 1,2,3,4$$
$$x_{ij} \ge 0,\ 0 \le (\alpha_{\widetilde{c}_{ij}})^3 + (\beta_{\widetilde{c}_{ij}})^3 \le 1,\ i = 1,2,3,\ j = 1,2,3,4$$

Utilizing Type 1 ($k = 1$), Type 2 ($k = 2$) and Type 3 ($k = 3$) score functions and Table 3 in LPP (3), we have obtained three LPP namely LPP (4), LPP (5) and LPP (6) which are as follows:

$$\text{minimize } S_{1F}\left\langle \alpha_{z_0}, \beta_{\widetilde{z}_0} \right\rangle = \sum_{i=1}^{3}\sum_{j=1}^{4} S_{1F}(\left\langle \alpha_{\widetilde{c}_{ij}}, \beta_{\widetilde{c}_{ij}} \right\rangle) \odot x_{ij} \tag{4}$$

subject to

$$\sum_{j=1}^{4} x_{ij} \le S_{1F}(\left\langle \alpha_{\widetilde{a}_i}, \beta_{\widetilde{a}_i} \right\rangle),\ i = 1,2,3$$
$$\sum_{i=1}^{3} x_{ij} \ge S_{1F}(\left\langle \alpha_{\widetilde{b}_j}, \beta_{\widetilde{b}_j} \right\rangle),\ j = 1,2,3,4$$
$$x_{ij} \ge 0,\ i = 1,2,3,\ j = 1,2,3,4$$

The score values corresponding to transportation cost, supply and demand of LPP (4) are given in Table 5.

$$\text{minimize } S_{2F}\left\langle \alpha_{z_0}, \beta_{\widetilde{z}_0} \right\rangle = \sum_{i=1}^{3}\sum_{j=1}^{4} S_{2F}(\left\langle \alpha_{\widetilde{c}_{ij}}, \beta_{\widetilde{c}_{ij}} \right\rangle) \odot x_{ij} \tag{5}$$

subject to

$$\sum_{j=1}^{4} x_{ij} \le S_{2F}(\left\langle \alpha_{\widetilde{a}_i}, \beta_{\widetilde{a}_i} \right\rangle),\ i = 1,2,3$$
$$\sum_{i=1}^{3} x_{ij} \ge S_{2F}(\left\langle \alpha_{\widetilde{b}_j}, \beta_{\widetilde{b}_j} \right\rangle),\ j = 1,2,3,4$$
$$x_{ij} \ge 0,\ i = 1,2,3,\ j = 1,2,3,4$$

Table 5
Score values corresponding to transportation cost, supply and demand of LPP (4).

	O1	O2	O3	O4	Supply
WH1	0.136	0.248	0.245	0.136	0.671
WH2	0.015	0.149	0.258	0.329	0.756
WH3	0.245	0.276	0.168	0.140	0.864
Demand	0.361	0.658	0.756	0.581	

Table 6
Score values corresponding transportation cost, supply and demand of LPP (5).

	O1	O2	O3	O4	Supply
WH1	0.0910	0.1680	0.1633	0.0910	0.5617
WH2	0.0099	0.1083	0.1807	0.2197	0.6743
WH3	0.1633	0.2053	0.1330	0.0957	0.8190
Demand	0.2617	0.5530	0.6743	0.4621	

Table 7
Score values corresponding to transportation cost, supply and demand when of LPP (6).

	O1	O2	O3	O4	Supply
WH1	0.0800	0.1200	0.1295	0.0800	0.4440
WH2	0.0098	0.0840	0.1125	0.1560	0.5705
WH3	0.1295	0.1040	0.0875	0.0805	0.7200
Demand	0.1005	0.2800	0.5705	0.1260	

The score values corresponding to transportation cost, supply and demand of LPP (5) are given in Table 6.

$$\text{minimize } S_{3F}\left\langle \alpha_{z_0}, \beta_{\tilde{z}_0} \right\rangle = \sum_{i=1}^{3}\sum_{j=1}^{4} S_{3F}(\left\langle \alpha_{\tilde{c}_{ij}}, \beta_{\tilde{c}_{ij}} \right\rangle) \odot x_{ij} \tag{6}$$

subject to

$$\sum_{j=1}^{4} x_{ij} \le S_{3F}(\left\langle \alpha_{\tilde{a}_i}, \beta_{\tilde{a}_i} \right\rangle),\ i = 1,2,3$$
$$\sum_{i=1}^{3} x_{ij} \ge S_{3F}(\left\langle \alpha_{\tilde{b}_j}, \beta_{\tilde{b}_j} \right\rangle),\ j = 1,2,3,4$$
$$x_{ij} \ge 0,\ i = 1,2,3\ j = 1,2,3,4$$

The score values corresponding to transportation cost, supply and demand of LPP (6) are given in Table 7.

It is to be noted that, the example considered here is numerically same with the example solved by Kumar et al. [53] but descriptively in different. Now, for comparison purpose, we have applied our proposed solution methodology to solve PFTP. So, to solve PFTP, we have re-considered all the input data of Table 3 as PFSs and consequently, the PFTP is as follows:

$$\text{minimize } S_P\left\langle \alpha_{\tilde{z}_0}, \beta_{\tilde{z}_0} \right\rangle = \sum_{i=1}^{3}\sum_{j=1}^{4} x_{ij} \odot S_P(\left\langle \alpha_{\tilde{c}_{ij}}, \beta_{\tilde{c}_{ij}} \right\rangle) \tag{7}$$

subject to

$$\sum_{j=1}^{4} x_{ij} \le S_P(\left\langle \alpha_{\tilde{a}_i}, \beta_{\tilde{a}_i} \right\rangle),\ i = 1,2,3$$
$$\sum_{i=1}^{3} x_{ij} \ge S_P(\left\langle \alpha_{\tilde{b}_j}, \beta_{\tilde{b}_j} \right\rangle),\ j = 1,2,3,4$$
$$\text{where } 0 \le \left(\alpha_{\tilde{z}_0}\right)^2 + \left(\beta_{\tilde{z}_0}\right)^2 \le 1$$
$$0 \le (\alpha_{\tilde{a}_i})^2 + (\beta_{\tilde{a}_i})^2 \le 1,\ i = 1,2,3$$
$$0 \le (\alpha_{\tilde{b}_j})^2 + (\beta_{\tilde{b}_j})^2 \le 1,\ j = 1,2,3,4$$
$$x_{ij} \ge 0,\ 0 \le (\alpha_{\tilde{c}_{ij}})^2 + (\beta_{\tilde{c}_{ij}})^2 \le 1,\ i = 1,2,3,\ j = 1,2,3,4$$

The score values corresponding to transportation cost, supply and demand of LPP (7) are given in Table 8.

It is worth mentioning that, we have taken same Table i.e., Table 3 for input data, but all the data viz. transportation cost, supply and demand satisfies Pythagorean fuzzy set property which is defined in Section 2 (see Definition 2.1).

Table 8
Score values corresponding to transportation cost, supply and demand of LPP (7).

	O1	O2	O3	O4	Supply
WH1	0.1000	0.2000	0.1850	0.1000	0.7400
WH2	0.0100	0.1400	0.2250	0.2600	0.8150
WH3	0.8150	0.2600	0.1750	0.1150	0.9000
Demand	0.3350	0.7000	0.8150	0.6050	

Table 9
Numerical results of LPP (4), LPP (5) and LPP (6).

Problem	Score function used	Optimal solutions	Minimum cost
LPP (4)	$\frac{1}{2}(1+\alpha^3_{\widetilde{F}}-\beta^3_{\widetilde{F}})$	$x_{12}=0.198,\ x_{14}=0.473$ $x_{21}=0.361,\ x_{22}=0.395$ $x_{33}=0.756,\ x_{34}=0.108$	$S_{1F}\left\langle\alpha_{\widetilde{z}_0},\beta_{\widetilde{z}_0}\right\rangle=0.31983$ $\left\langle\alpha_{\widetilde{z}_0},\beta_{\widetilde{z}_0}\right\rangle=\langle 0.39526, 0.79431\rangle$
LPP (5)	$\frac{1}{3}(1+2\alpha^3_{\widetilde{F}}-\beta^3_{\widetilde{F}})$	$x_{12}=0.1404,\ x_{14}=0.4213,$ $x_{21}=0.2617,\ x_{22}=0.4126,$ $x_{33}=0.6743,\ x_{34}=0.0408$	$S_{2F}\left\langle\alpha_{\widetilde{z}_0},\beta_{\widetilde{z}_0}\right\rangle=0.20279$ $\left\langle\alpha_{\widetilde{z}_0},\beta_{\widetilde{z}_0}\right\rangle=\langle 0.38228, 0.82105\rangle$
LPP (6)	$\frac{1}{2}(1+\alpha^2_{\widetilde{F}}-\beta^2_{\widetilde{F}})\left\lvert\alpha_{\widetilde{F}}-\beta_{\widetilde{F}}\right\rvert$	$x_{14}=0.1260,\ x_{21}=0.1005,$ $x_{22}=0.2800,\ x_{33}=0.5705$	$S_{3F}\left\langle\alpha_{\widetilde{z}_0},\beta_{\widetilde{z}_0}\right\rangle=0.084504$ $\left\langle\alpha_{\widetilde{z}_0},\beta_{\widetilde{z}_0}\right\rangle=\langle 0.35437, 0.90132\rangle$

Table 10
Comparison results.

Method	Problem	Score function used	Optimal solutions	Minimum cost
Our proposed method	LPP (7) in this paper	$\frac{1}{2}(1+\alpha^2_{\widetilde{F}}-\beta^2_{\widetilde{F}})$	$x_{12}=0.220$ $x_{14}=0.520$ $x_{21}=0.335$ $x_{22}=0.480$ $x_{33}=0.815$ $x_{34}=0.085$	$\left\langle\alpha_{\widetilde{z}_0},\beta_{\widetilde{z}_0}\right\rangle=\langle 0.43060, 0.77673\rangle$ $S_P\left\langle\alpha_{\widetilde{z}_0},\beta_{\widetilde{z}_0}\right\rangle=0.31895$
Method proposed by Kumar et al. [27]	Example 5.3 of Kumar et al. [27]	$\frac{1}{2}(1+\alpha^2_{\widetilde{F}}-\beta^2_{\widetilde{F}})$	$x_{12}=0.220$ $x_{14}=0.520$ $x_{21}=0.335$ $x_{22}=0.48$ $x_{33}=0.815$ $x_{34}=0.085$	$S_P\left\langle\alpha_{\widetilde{z}_0},\beta_{\widetilde{z}_0}\right\rangle=0.31895$

LPP (4), LPP (5) and LPP (6) have been solved using excel solver and numerical results have been presented in Table 9. As no researchers have solved TP considering Fermatean Fuzzy parameters, therefore no comparisons have been made for the same. Again, LPP (7) has already been considered in the paper published by Kumar et al. [53]. Therefore, we have been solved LPP (7) using for the purpose of comparison of our proposed method. The comparisons results have been shown in Table 10. From Table 9, it is observed that Fermatean fuzzy cost $\langle 0.35437, 0.90132\rangle$ is less than $\langle 0.38228, 0.82105\rangle$ and $\langle 0.39526, 0.79431\rangle$. Also, $\langle 0.38228, 0.82105\rangle$ is less than $\langle 0.39526, 0.79431\rangle$.

So, Type 3 sore function is better than Type 2 score function and Type 2 score function is better than Type 1 score functions. From Table 9, it is seen that, the minimum cost obtained by Kumar et al. [53] using score function of Pythagorean fuzzy set is 0.31896. Again, from Table 10, it is also seen that, using same score function, we get same result by using our proposed method. Therefore, we claimed that our proposed solution method is an alternative method for solving transportation problem in Fermatean fuzzy environment. Further, it is noticed from Table 10, that Kumar et al. [53] have obtained minimum cost in terms of precise valued but we have obtained minimum transportation cost in terms of precise valued as well as Fermatean fuzzy valued. Therefore, we may claim that our proposed method is a new way to tackle the uncertainty in Fermatean fuzzy environment.

6. Concluding remark

In this paper, first time, a method is proposed for finding the optimal solution of fuzzy transportation problems in which parameters relating to the transportation problem are Fermatean fuzzy sets. Transportation problem is one of the most important problems in decision making. In many real life situations, costs, supply and demands of the transportation problem are not precise. The transportation problem with Fermatean fuzzy parameters is more realistic than the transportation problem with precise parameters because most of the real life situations are comes from uncertain domain. The existing arithmetic operations of the Fermatean fuzzy sets are employed to find optimum solution. Here, we have proposed new definition of score function for FFS and for any Fermatean fuzzy set $\widetilde{F}$, the sore function value lies in the unit interval i.e., $0 \le S_F(\widetilde{F}) \le 1$. From, computational results, it is

mentioned that the FFSs have more appropriate than PFSs and are capable to manage higher level of uncertainties. The proposed methodology has been discussed with a numerical example. The proposed method discussed here is very simple and very easy to implement and to apply for solving realistic decision making problems in the areas like management science, engineering, operations research and others decision making subjects occurring in the near future involving Fermatean fuzzy parameters.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The author is very much grateful to Editor-in-Chief and anonymous reviewers for their valuable and constructive suggestions to improve the entire paper.

References

[1] Tirkolaee EB, Goli A, Weber GW. Fuzzy mathematical programming and self-adaptive artificial fish swarm algorithm for just-in-time energy-aware flow shop scheduling problem with outsourcing option. IEEE Trans Fuzzy Syst 2020;28(11):2772–83.
[2] Tirkolaee EB, Goli A, Pahlevan M, Kordestanizadeh RM. A robust bi-objective multi-trip periodic capacitated arc routing problem for urban waste collection using a multi-objective invasive weed optimization. Waste Manage Res 2019;37(11):1089–101.
[3] Tirkolaee EB, Mahdavi I, Esfahani MMS, Weber GW. A hybrid augmented ant colony optimization for the multi-trip capacitated arc routing problem under fuzzy demands for urban solid waste management. Waste Manage Res 2019;38(2):156–72.
[4] Tirkolaee EB, Goli A, Weber GW. Multi-objective aggregate production planning model considering overtime and out sourcing options under fuzzy seasonal demand. In: Hamrol A, Kujawinska A, Barraza M, editors. Advances in manufacturing II. Manufacturing. Lecture notes in mechanical engineering, Cham: Springer; 2019, http://dx.doi.org/10.1007/978-030-18789-7_8.
[5] Tirkolaee EB, Goli A, Bakhsi M, Mahdavi I. A robust multi-trip vehicle routing problem of perishable products with intermediate depots and time windows. Numer Algebra Control Optim 2017;7(4):417–33.
[6] Tirkolaee EB, Abbasian P, Weber GW. Sustainable fuzzy multi-trip location-routing problem for medical waste management during the COVID-19 outbreak. Sci Total Environ 2021;756:143607.
[7] Barman H, Pervin M, Roy SK, Weber GW. Back-ordered inventory model with inflation in a cloudy-fuzzy environment. J Ind Manage Optim 2021;17(4):1913–41.
[8] Goli A, Tirkolaee EB, Aydin NS. Fuzzy integrated cell formation and production scheduling considering automated guided vehicles and human factors. IEEE Trans Fuzzy Syst 2021. http://dx.doi.org/10.1109/TFUZZ.2021.3053838, 2021.
[9] Ozceylan E, Paksoy T. Interactive fuzzy programming approaches to the strategic and tactical planning of a closed-loop supply chain under uncertainty. Int J Prod Res 2014;52(8):2363–87.
[10] Paksoy T, Pehlivan NY, Ozceylan E. Fuzzy multi-objective optimization of a green supply chain network with risk management that includes environmental hazards. Hum Ecol Risk Assess 2012;18(5):1120–51.
[11] Paksoy T, Gungor A, Ozceylan E, Hancilar A. Mixed model disassembly line balancing problem with fuzzy goals. Int J Prod Res 2013;51(20):6082–96.
[12] Demirel N, Ozceylan E, Paksoy T, Gokcen H. A genetic algorithm approach for optimising a closed-loop supply chain network with crisp and fuzzy objectives. Int J Prod Res 2014;52(12):3637–64.
[13] Bhaumik A, Roy SK, Weber GW. Multi-objective linguistic-neutrosophic matrix game and its applications to tourism management. J Dyn Games 2021;8(2):101–18.
[14] Lotfi R, Mardani N, Weber GW. Roubst bi-level programming for renewable energy location. Int J Energy Res 2021;45(2):1–15.
[15] Hitchcock FL. The distribution of product from several resources to numerous localities. J Math Phys 1941;20:224–30.
[16] Korukoglu S, Balli S. A improved Vogel's approximation method for the transportation problem. Math Comput Appl 2011;16:370–81.
[17] Zadeh LA. Fuzzy sets. Inf Control 1965;8:338–53.
[18] Bellman RE, Zadeh LA. Decision making in a fuzzy environment. Manage Sci 1970;17:141–64.
[19] H.J. Zimmermann. Fuzzy programming and linear programming with several objective functions. Fuzzy Sets and Systems 1978;1:45–55.
[20] Chanas S, Kolodziejczyk W, Machaj A. A fuzzy approach to the transportation problem. Fuzzy Sets and Systems 1984;13:211–21.
[21] Chanas S, Kuchata D. A concept of the optimal solution of the transportation problem with fuzzy cost coefficients. Fuzzy Sets and Systems 1996;82(3):299–305.
[22] Tada M, Ishii H. An integer fuzzy transportation problem. Comput Math Appl 1996;31:71–87.
[23] Hashmi N, Jalil SA, Javaid S. A model for two stage fixed charge transportation problem with multiple objectives and fuzzy linguistic preferences. Soft Comput 2019;23:12401–15.
[24] Liu P, Yang L, Wang L, Li S. A solid transportation problem with type-2 fuzzy variables. Appl Soft Comput 2014;24:543–58.
[25] Kundu P, Kar S, Maiti M. Fixed charge transportation problem with type-2 fuzzy variables. Inform Sci 2014;255:170–86.
[26] Singh SK, Yadav SP. A new approach for solving intuitionistic fuzzy transportation problem of type-2. Ann Oper Res 2016;243:349–63.
[27] Gupta G, Kumari A. An efficient method for solving intuitionistic fuzzy transportation problem of type-2. Int J Appl Comput Math 2017;3:3795–804.
[28] Arora J. An algorithm for interval-valued fuzzy fractional transportation problem. SKIT Res J 2018;8:71–5.
[29] Bharti SK, Singh SR. Transportation problem under interval-valued intuitionistic fuzzy environment. Int J Fuzzy Syst 2018;20:1511–22.
[30] Kumar PS. PSK method for solving intuitionistic fuzzy solid transportation problems. Int J Fuzzy Syst Appl 2018;7(4):62–99.
[31] Kaur A, Kumar A. A new method for solving fuzzy transportation problems using ranking function. Appl Math Model 2011;35:5652–61.
[32] Midya S, Roy SK, Yu VF. Intuitionistic fuzzy multi-stage multi-objective fixed-charge solid transportation problem in green supply chain. Int J Mach Learn Cybern 2021;12:699–717.
[33] Ghosh S, Roy SK. Fuzzy-rough multi-objective product blending fixed-charge transportation problem with truck load constraints through transfer station. RAIRO Oper Res 2021;55:S2923–52.
[34] Ghosh S, Roy SK, Ebrahimnejad A, Verdegay JL. Multi-objective fully intuitionistic fuzzy fixed-charge solid transportation problem. Complex Intell Syst 2021;7(2):1009–23.
[35] Das SK, Roy SK, Weber GW. Application of type-2 fuzzy logic to a multi-objective green solid transportation-location problem with dwell time under carbon tax, cap and offset policy: Fuzzy versus non-fuzzy techniques. IEEE Trans Fuzzy Syst 2020;28(11):2711–25.

[36] Das SK, Roy SK. Effect of variable carbon emission in a multi-objective transportation-p-facility location problem under neutrosophic environment. Comput Ind Eng 2019;132:311–24.
[37] Roy SK, Midya S, Weber GW. Multi-objective fixed-charge transportation problem under two-fold uncertainty. Neural Comput Appl 2019;31:8593–613.
[38] Ahmad F, Adhami AY. Neutrosophic programming approach to multiobjective nonlinear transportation problem with fuzzy parameters. Int J Manage Sci 2019;14(3):218–29.
[39] Li L, Lai KK. A fuzzy approach to the multiobjective transportation problem. Comput Oper Res 2000;27:43–57.
[40] Atanassov KT. Intuitionistic fuzzy sets. Fuzzy Sets and Systems 1986;20:87–96.
[41] Atanassov KT. On intuitionistic fuzzy sets theory. Berlin: Springer; 2012.
[42] Atanassov KT, Vassilev PM, Tsvetkov RT. Intuitionistic fuzzy sets, measures and integrals. Sofia: Professor Marin Drinov Academic publishing House; 2013.
[43] Atanassov KT. A second type of intuitionistic fuzzy sets. BUSE-FAL 1983;56:66–70.
[44] Atanassov KT. Geometric interpretation of the elements of the intuitionistic fuzzy objects. Int J Bioautom 2016;20(S1):S27–42.
[45] Yager RR. Pythagorean fuzzy subsets. In: 2013 joint IFSA world congress and NAFIPS annual meeting (IFSA/ NAFIPS). 2013, p. 57–61.
[46] Yager RR. Pythagorean membership grades in multicriteria decision making. IEEE Trans Fuzzy Syst 2014;22:958–65.
[47] Li Z, Wei G, Lu M. Pythagorean fuzzy hamy mean operators in multiple attribute group decision making and their application to supplier selection. Symmetry 2018;10:505.
[48] Zhou J, Su W, Balezentis T, Streimikiene D. Multiple criteria group decision making considering symmetry with regards to the positive and negative ideal solution via the Pythagorean normal cloud model for application to economic decision. Symmetry 2018;10(5):140.
[49] Boltruk E. Pythagorean fuzzy CODAS and its application to supplier selection in a manufacturing firm. J Enterp Inf Manage 2018;31:550–64.
[50] Qin J. Generalized Pythagorean fuzzy maclaurin symmetric means and its application to multiple attribute sir group decision model. Int J Fuzzy Syst 2018;20:943–57.
[51] Wan S-P, Li S-Q, Dong J-Y. A three phase method for Pythagorean fuzzy multi attribute group decision making and application to haze management. Comput Ind Eng 2018;123:348–63.
[52] Chen T-Y. An outranking approach using a risk attitudinal assignment model involving Pythagorean fuzzy information and its application to financial decision making. Appl Soft Comput 2018;71:460–87.
[53] Kumar R, Edalatpanah SA, Jha S, Singh R. A Pythagorean fuzzy approach to the transportation problem. Complex Intell Syst 2019;5:255–63.
[54] Senapati T, Yager RR. Fermatean fuzzy sets. J Ambient Intell Humaniz Comput 2020;11:663–74.
[55] Senapati T, Yager RR. Fermatean fuzzy weighted averaging/geometric operators and its application in multi-criteria decision-making methods. Eng Appl Artif Intell 2019;85:112–21.
[56] Liu ST, Kao C. Solving fuzzy transportation problems based on extension principle. Eur J Oper Res 2004;153. 661-674.
[57] Dinagar DS, Palanivel K. The transportation problem in fuzzy environment. Int J Algorithms Comput Math 2009;2:65–71.
[58] Pandian P, Natarajan G. A new algorithm for finding a fuzzy optimal solution for fuzzy transportation problems. Appl Math Sci 2010;4:79–90.
[59] Azevedo N, Pinheiro D, Weber GW. Dynamic programming for a Markov-switching jump-diffusion. J Comput Appl Math 2014;267:1–19.
[60] Savku E, Azevedo N, Weber GW. Optimal control of stochastic hybrid models in the framework of regime switches. In: International conference on dynamics, games and science. Springer; 2014, p. 371–87.
[61] Garg H. Anew improved score function of an interval valued Pythagorean fuzzy set based topsis method. Int J Uncertain Quantif 2017;7:463–74.
[62] Geng Y, Liu P, Teng F, Liu Z. Pythagorean fuzzy uncertain linguistic TODIM method and their application to multiple criteria group decision making. Int J Intell Syst 2017;33:3383–95.
[63] Garg H. A linear programming method based on an improved score function for interval valued Pythagorean fuzzy numbers and its application to decision-making. Int J Uncertain Fuzz 2018;26:67–80.
[64] Kropat E, Weber GW. Fuzzy target-environment networks and fuzzy-regression approaches. Numer Algebra Control Optim 2018;8(2):135–55.
[65] Sahoo L. Some score functions on fermatean fuzzy sets and its application to bride selection based on TOPSIS method. Int J Fuzzy Syst Appl 2021;10(3):18–29.